

Point Plot Analysis: Board & Gender Trends

Seaborn Visualization Analysis & Summary Report

Visualization Overview

This report analyzes the point plot showing estimated mean values for **Secondary School Percentage (ssc_p)** grouped by **Higher Secondary Education Board (hsc_b)** and hue-split by **Gender (M vs F)**.

Summary Table of Estimates

Board (hsc_b)	Gender	Mean ssc_p Score	95% Confidence Interval
Central	Male (M)	~65.8%	~63.0% to ~68.0%
Central	Female (F)	~67.4%	~64.0% to ~70.6%
Others	Male (M)	~67.4%	~65.0% to ~69.8%
Others	Female (F)	~69.0%	~65.0% to ~72.8%

Key Observations

- Gender Advantage:** Female students consistently outperform male students across both board types by approximately **1.6 percentage points**.
- Board Differences:** Students in the "Others" board average higher overall scores (~67.4% for males, ~69.0% for females) compared to the "Central" board (~65.8% for males, ~67.4% for females).
- Interaction Effect:** Parallel trendlines indicate no significant statistical interaction between gender performance and high school board selection.